

PATENT APPLICATION

System and Method for AI-Powered Asset Tokenization on a Custom Blockchain Platform with Integrated Escrow, Trading Engine, and Algorithmic Price Discovery

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ABSTRACT

The present invention relates to a comprehensive system and method for real-world asset tokenization using a custom-built blockchain infrastructure integrated with artificial intelligence-powered document verification, an automated escrow mechanism, a centralized limit-order trading engine, and a dynamic algorithmic price discovery model. The system enables asset owners to create digital tokens representing fractional ownership of real-world assets through a multi-stage lifecycle that includes AI-driven document analysis, compliance review, and on-chain tokenization. The platform employs a custom proof-of-work blockchain utilizing SHA-256 hashing and ECDSA secp256k1 cryptographic signatures, supporting eight distinct transaction types including MINT, TRANSFER, INVEST, DIVEST, PAYOUT, REFUND, FEE, and ASSET_CREATE. The trading engine implements price-time priority order matching with atomic settlement recorded on the blockchain ledger. The price discovery model incorporates both supply-based and asset-funding-based adjustment factors, computing a dynamic token price that reflects platform economic activity. The system further includes a multi-tiered security model comprising JWT-based authentication, token-bucket rate limiting, input sanitization, audit logging with hash-chain integrity verification, and double-mint prevention through atomic database operations. The invention finds particular application in the fields of financial technology, decentralized finance (DeFi), and regulatory-compliant digital asset management.

FIELD OF THE INVENTION

This invention relates to the field of financial technology (fintech), and more particularly to systems, methods, and computer-readable media for the tokenization of real-world assets on a blockchain-based platform. Specifically, the invention encompasses the integration of artificial intelligence for automated asset verification, a custom proof-of-work blockchain for immutable transaction recording, an escrow-based investment mechanism for capital raising, a centralized order-matching trading engine for secondary market liquidity, and an algorithmic pricing model for dynamic token valuation. The invention addresses the technical challenges of combining on-chain immutability with off-chain compliance requirements, thereby enabling regulated asset tokenization in jurisdictions requiring documented verification of underlying assets.

The intersection of blockchain technology and traditional financial instruments has given rise to a new paradigm known as real-world asset (RWA) tokenization. This paradigm involves the creation of digital tokens on a distributed ledger that represent ownership rights, economic benefits, or other entitlements linked to physical or financial assets such as real estate, commodities, receivables, or equity stakes in private companies. The technical challenges in this domain are multifaceted, spanning cryptographic security, regulatory compliance, market microstructure design, and user experience. The present invention provides an integrated solution that addresses these challenges within a single unified platform architecture, thereby advancing the state of the art in asset tokenization technology.

BACKGROUND OF THE INVENTION

3.1 Conventional Asset Tokenization Approaches

Existing asset tokenization platforms typically operate on public blockchain networks such as Ethereum, Binance Smart Chain, or Polygon, utilizing smart contracts to mint and manage tokenized representations of real-world assets. These approaches suffer from several technical limitations. First, reliance on public blockchains introduces scalability constraints, as transaction throughput is limited by the blockchain's consensus mechanism, often resulting in high gas fees and delayed confirmation times that are unsuitable for high-frequency trading operations. Second, public blockchains expose transaction details to all network participants, creating privacy concerns for asset owners who may not wish to disclose investment amounts, investor identities, or asset valuations to the public. Third, the open-ended nature of smart contract code on public networks increases the attack surface for vulnerabilities, as demonstrated by numerous high-profile exploits in the decentralized finance (DeFi) space.

Alternative approaches have employed permissioned or private blockchain networks to address privacy and scalability concerns. However, these solutions typically lack the native trading infrastructure, algorithmic price discovery, and integrated escrow mechanisms necessary for a complete asset tokenization lifecycle. They often require external integration with centralized exchanges or over-the-counter (OTC) markets for secondary trading, introducing counterparty risk and operational complexity. Furthermore, existing systems generally lack automated, AI-driven verification of asset documentation, relying instead on manual review processes that are slow, expensive, and prone to human error or bias.

3.2 Limitations of Prior Art

Several technical deficiencies exist in the prior art that the present invention seeks to overcome. First, there is no known system that integrates a purpose-built proof-of-work blockchain with a five-stage AI document analysis pipeline, an automated escrow mechanism for capital raising, and a price-time priority trading engine within a single unified architecture. Existing systems treat these as separate components requiring manual orchestration, leading to data inconsistency, increased latency, and a fragmented user experience.

Second, existing price discovery mechanisms for tokenized assets are typically limited to simple supply-demand curves or oracle-based external price feeds. These approaches fail to account for the unique economic dynamics of a tokenization platform, where the relationship between token supply, asset funding activity, and market liquidity creates complex feedback loops that require a more sophisticated pricing model. The present invention introduces a dual-factor pricing formula that incorporates both supply-based inflation and asset-funding-based demand signals, providing a more accurate and responsive price discovery mechanism.

Third, the integration of an on-chain escrow system with an asset lifecycle state machine represents a novel technical contribution. In existing systems, escrow functionality, when present, is implemented as an external smart contract or third-party service, leading to additional trust assumptions, transaction costs, and potential points of failure. The present invention embeds the escrow mechanism directly within the platform's transaction type system, enabling atomic INVEST, DIVEST, PAYOUT, and REFUND operations that are recorded immutably on the blockchain ledger.

SUMMARY OF THE INVENTION

The present invention provides a comprehensive system and method for real-world asset tokenization that overcomes the limitations of prior art by integrating five core technical components into a unified platform architecture. These components are: (i) a custom proof-of-work blockchain engine with eight specialized transaction types, (ii) a five-stage artificial intelligence pipeline for automated asset document verification, (iii) an on-chain escrow mechanism with automated payout and refund logic, (iv) a centralized order-matching trading engine with price-time priority execution, and (v) a dual-factor algorithmic price discovery model.

The blockchain engine employs SHA-256 proof-of-work mining with dynamic difficulty adjustment and stores the complete chain in a structured JSON format (chain.json). Each block contains a Merkle root computed from all included transactions, a reference to the previous block's hash, a nonce value determined through the mining process, and a difficulty parameter governing the mining complexity. Transactions are signed using ECDSA secp256k1 elliptic curve cryptography, providing a 128-bit security level that is computationally infeasible to break with current or foreseeable computing technology. The system supports eight distinct transaction types (MINT, TRANSFER, INVEST, DIVEST, PAYOUT, REFUND, FEE, and ASSET_CREATE), each serving a specific function within the asset tokenization lifecycle.

The AI document analysis pipeline processes uploaded asset documents through five sequential stages: image preprocessing and enhancement, optical character recognition (OCR) for text extraction from scanned documents, text classification and categorization, key information extraction using named entity recognition, and a final verification scoring stage that produces a confidence score and recommendation (verified or rejected). This pipeline enables the automated verification of property deeds, financial statements, valuation reports, and other supporting documentation without requiring manual review, thereby significantly reducing the time and cost associated with asset onboarding.

The escrow mechanism operates on a per-asset basis, with each tokenized asset having a dedicated escrow account identified by a unique blockchain address. When investors purchase asset tokens, the corresponding funds are locked in the escrow account through an atomic INVEST transaction recorded on the blockchain. Upon successful funding of the asset, the escrow releases the total locked amount minus a platform fee (typically 1.0% of the raised amount) to the asset owner via a PAYOUT transaction. If the asset fails to reach its funding target within the specified duration, the escrow automatically processes REFUND transactions to return funds to all investors. This mechanism eliminates counterparty risk and ensures that investor funds are protected throughout the funding period.

The trading engine implements a centralized limit-order book with price-time priority matching. Users can place both market orders (executed immediately at the best available price) and limit orders (executed only at or better than a specified price). The engine matches buy and sell orders based on price priority (best price first) and time priority (earliest order at the same price level first), following the standard price-time priority discipline used in established financial exchanges. Each executed trade is recorded on the blockchain as a TRADE transaction, providing an immutable audit trail of all secondary market activity. The engine enforces balance verification against the blockchain before order placement, preventing the creation of orders that exceed the user's available token balance.

The algorithmic price discovery model computes the platform token price using a dual-factor formula: $P = P_{\text{initial}} \times (1 + \text{totalSupply}/10000) \times (1 + \text{totalAssetsFunded} \times 0.04)$, where P_{initial} is the base token price, `totalSupply` represents the cumulative number of tokens minted on the platform, and `totalAssetsFunded` represents the cumulative number of assets that have successfully completed their funding period. This formula creates a positive feedback loop that aligns token value with platform activity: as more tokens are minted (indicating user adoption) and more assets are funded (indicating platform utility), the token price appreciates, incentivizing further participation in the ecosystem.

DETAILED DESCRIPTION OF THE INVENTION

5.1 System Architecture Overview

The system architecture of the present invention is organized into five principal layers: the Blockchain Layer, the Application Layer, the Data Layer, the AI Processing Layer, and the Security Layer. The Blockchain Layer provides the immutable ledger infrastructure and includes the custom proof-of-work mining engine, transaction pool management, and chain storage subsystem. The Application Layer exposes RESTful API endpoints through an Express.js server and implements the business logic for asset management, trading, payment processing, and user administration. The Data Layer utilizes a SQLite relational database with nineteen interconnected tables that maintain the platform's state, including user accounts, wallets, assets, orders, trades, escrow records, and economic metrics. The AI Processing Layer implements the five-stage document analysis pipeline for automated asset verification. The Security Layer implements authentication, authorization, rate limiting, input sanitization, and audit logging middleware that processes all incoming HTTP requests before they reach the application logic.

The system is deployed as a monolithic Node.js application with integrated blockchain mining, trading engine, and AI processing capabilities. While the monolithic architecture simplifies deployment and reduces inter-service communication overhead, the system is designed with clear module boundaries that would facilitate future migration to a microservices architecture if scalability requirements demand it. The blockchain storage is maintained in a JSON file (`chain.json`) that is periodically persisted to disk, while the relational data is stored in a SQLite database file that supports atomic transactions and referential integrity constraints.

5.2 Blockchain Engine

The blockchain engine is the foundational component of the system, providing an immutable, chronologically ordered record of all platform transactions. Unlike conventional blockchain implementations that rely on existing networks such as Ethereum or Bitcoin, the present invention implements a purpose-built blockchain that is tightly integrated with the platform's business logic. This design choice enables the definition of custom transaction types that directly correspond to platform operations, eliminating the need for smart contract interpretation layers and reducing transaction processing latency.

Transaction Type	Code	Description
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MINT	Coin creation from fiat purchase	Creates new platform tokens when a user purchases tokens through a fiat payment gateway
TRANSFER	User-to-user or user-to-system transfer	Moves tokens between wallet addresses, including platform fee collections
INVEST	User to Escrow (buying asset tokens)	Locks investor funds in the asset's escrow account and records the investment on chain
DIVEST	Exit from investment	Processes early withdrawal from an active investment, returning funds to the investor
PAYOUT	Escrow to Owner (asset fully funded)	Releases escrowed funds to the asset owner upon successful completion of the funding period
REFUND	Escrow to Investors (asset expired)	Returns escrowed funds to all investors when an asset fails to reach its funding target
FEE	Fee collection	Records the collection of trading, listing, capital raise, and withdrawal fees
ASSET_CREATE	Asset tokenization recorded on chain	Immutably records the creation of a new tokenized asset on the blockchain ledger

Table 1: Transaction Types Supported by the Custom Blockchain Engine

Each transaction includes a cryptographic signature generated using the ECDSA secp256k1 elliptic curve algorithm, which is the same cryptographic scheme employed by Bitcoin. The secp256k1 curve provides a security level of approximately 128 bits, meaning that an attacker would need to perform approximately 2^{128} operations to forge a signature, which is computationally infeasible with current and foreseeable computing technology. Each transaction also includes a SHA-256 hash of the transaction data, which serves as a unique identifier and is used in the Merkle root computation for block verification.

The proof-of-work mining process involves finding a nonce value such that the SHA-256 hash of the block header (which includes the previous block hash, Merkle root, timestamp, difficulty target, and nonce) is below the difficulty target. The difficulty is dynamically adjusted based on the network's mining rate to maintain a consistent block generation interval. This mechanism ensures that the blockchain remains resistant to tampering, as any modification to a historical block would require re-mining that block and all subsequent blocks, a task that becomes exponentially more difficult as the chain grows.

5.3 AI-Powered Document Analysis Pipeline

The AI document analysis pipeline represents a key innovative component of the present invention, enabling the automated verification of asset documentation that is critical for regulatory compliance and investor protection. The pipeline processes user-uploaded documents (supporting JPG, PNG, WebP, and PDF formats with a maximum file size of 10MB each) through five sequential stages, each designed to extract progressively deeper levels of information from the document content.

The first stage, image preprocessing and enhancement, applies standard computer vision techniques including noise reduction, contrast enhancement, skew correction, and binarization to prepare the

document image for subsequent analysis. This stage is particularly important for scanned documents or photographs that may suffer from poor lighting, shadow artifacts, or perspective distortion. The second stage, optical character recognition (OCR), extracts raw text content from the preprocessed image using a deep learning-based OCR engine capable of handling multiple languages, fonts, and layout formats commonly encountered in legal and financial documents.

The third stage, text classification and categorization, analyzes the extracted text to determine the document type (e.g., property deed, financial statement, valuation report, identity document) and assesses its relevance to the asset being tokenized. This stage employs natural language processing (NLP) techniques including text classification models trained on domain-specific corpora of financial and legal documents. The fourth stage, key information extraction, uses named entity recognition (NER) to identify and extract structured data points from the unstructured text, including asset addresses, monetary amounts, dates, legal entity names, registration numbers, and other relevant information. The fifth and final stage, verification scoring, aggregates the outputs of all preceding stages to produce a confidence score (0-100) and a binary recommendation (verified or rejected) that determines whether the asset proceeds to compliance review or is returned to the user for remediation.

5.4 Asset Lifecycle State Machine

The asset lifecycle is managed through a deterministic finite state machine that governs the valid transitions between asset states. The state machine enforces business rules and regulatory requirements by restricting state transitions to predefined paths, preventing invalid operations and ensuring that each asset follows the correct verification and funding process. The complete state transition sequence is: draft, documents_uploaded, ai_analyzing, verified (or rejected), compliance_review, active, funding, funded (or expired), payout_pending, completed (or refunding, closed).

In the draft state, the asset owner provides essential metadata including the asset title, description, category (selected from twelve predefined categories), target raise amount (bounded between 100 INR and 1 Crore INR), and listing duration. Upon validation, the asset is inserted into the database with a draft status. In the documents_uploaded state, the owner uploads between one and ten supporting documents, which are stored in a dedicated directory with cryptographically randomized filenames to prevent unauthorized access. SHA-256 hashes are computed for each uploaded document to enable duplicate detection and integrity verification. The ai_analyzing state triggers the five-stage AI pipeline described in Section 5.3, which produces a verification score and recommendation.

Upon successful AI verification, the asset enters the compliance_review state, where it undergoes additional regulatory compliance checks. If approved, the asset transitions to the active state and becomes visible to investors on the platform. During the funding state, investors can purchase asset tokens through INVEST transactions that lock their funds in the asset's escrow account. If the cumulative investment reaches or exceeds the target raise amount before the listing duration expires, the asset transitions to the funded state, and subsequently to payout_pending and completed states as the escrow releases funds to the asset owner. If the funding period expires without reaching the target, the asset transitions to the expired state, triggering automatic REFUND transactions to return all investor funds.

5.5 Trading Engine and Order Matching

The trading engine provides secondary market liquidity for platform tokens through a centralized limit-order book architecture. The engine supports both market orders (which execute immediately at the best available price) and limit orders (which execute only at or better than a specified price). The order matching algorithm follows the price-time priority discipline: buy orders are sorted by price in descending order (highest bid first) and sell orders are sorted by price in ascending order (lowest ask first), with ties broken by timestamp (earliest order first). This matching discipline ensures fair and efficient price discovery consistent with the practices of established financial exchanges worldwide.

Before placing an order, the trading engine verifies the user's token balance against the blockchain ledger to ensure that the user has sufficient funds to cover the order. For sell orders, the engine checks that the user's wallet balance is greater than or equal to the order amount. For buy orders, the engine checks that the user's wallet balance is sufficient to cover the total cost (amount multiplied by price). This balance verification step prevents the creation of unfunded or partially funded orders that could lead to settlement failures.

Each executed trade is recorded on the blockchain as a TRADE transaction, which includes the buyer address, seller address, trade amount, execution price, buyer fee (0.1% of trade value), seller fee (0.1% of trade value), and the transaction hash. This on-chain settlement ensures that all secondary market activity is immutably recorded and publicly auditable. The trading engine also implements a double-mint prevention mechanism using atomic database operations: the payment verification step uses an atomic UPDATE query with a WHERE clause that locks the order record, preventing concurrent requests from processing the same order multiple times.

5.6 Algorithmic Price Discovery Model

The token price on the platform is determined by a dual-factor algorithmic formula rather than by external market forces or oracle price feeds. The price is computed as: $P = P_{\text{initial}} \times (1 + \text{totalSupply}/10000) \times (1 + \text{totalAssetsFunded} \times 0.04)$, where P_{initial} is the base token price set at platform initialization, totalSupply is the cumulative number of tokens in circulation, and totalAssetsFunded is the cumulative count of assets that have successfully completed their funding period. This formula has two multiplicative adjustment factors.

The first factor, $(1 + \text{totalSupply}/10000)$, introduces a supply-based inflation component. As more tokens are minted through MINT transactions (indicating user adoption and platform growth), the denominator of 10,000 ensures that the inflation rate remains manageable: at 100,000 tokens in circulation, the supply adjustment factor is 11.0, representing a 10x increase from the base price. This gradual appreciation incentivizes early platform participation while maintaining a predictable relationship between supply and price.

The second factor, $(1 + \text{totalAssetsFunded} \times 0.04)$, introduces a demand-based component tied to platform utility. As more assets are successfully funded (indicating that the platform is delivering real value by connecting asset owners with investors), the token price appreciates further. The coefficient of 0.04 (4% per funded asset) creates a meaningful but not excessive price impact per asset, ensuring that the price remains stable during periods of low funding activity while appreciating during periods of high platform utility.

5.7 Fee Structure and Economic Model

The platform implements a multi-tiered fee structure designed to align the economic incentives of all participants while generating sustainable revenue for platform operations. The fee structure comprises five fee types: trading fees (0.1% per trade, charged to both buyer and seller), asset listing fees (1.0 AC per asset, configurable but not currently enforced), capital raise fees (1.0% of the total raised amount, deducted from the escrow payout to the asset owner), withdrawal fees (0.5 AC per withdrawal, sent to the platform fee wallet), and payment gateway fees (variable, deducted from the fiat amount by the payment processor). All fees are recorded in a dedicated fee_ledger table and aggregated in the economy table for financial reporting and analytics purposes.

Fee Type	Rate	Collection Method
Trading Fee	0.1% per trade (both sides)	Deducted at trade execution, sent to platform fee wallet
Asset Listing Fee	1.0 AC per asset	Configured in system, not currently enforced
Capital Raise Fee	1.0% of raised amount	Deducted from escrow payout to asset owner
Withdrawal Fee	0.5 AC per withdrawal	Sent to platform fee wallet
Gateway Fee	Varies by provider	Deducted from fiat amount by payment processor

Table 2: Platform Fee Structure

5.8 Security Architecture

The security architecture of the present invention implements a defense-in-depth strategy with multiple overlapping security mechanisms operating at different layers of the system stack. The middleware pipeline processes every incoming HTTP request through a sequence of security checks before the request reaches the application logic. The pipeline consists of the following components in order: JSON body parsing with a 5MB payload limit, input sanitization to remove potentially malicious content, general rate limiting (100 requests per 60 seconds per IP address), and audit logging middleware that records a cryptographic hash chain of all API requests for tamper-evident audit trails.

Authentication is implemented using JSON Web Tokens (JWT) with Bearer token scheme. The system provides three authentication middleware variants: authenticate (which requires a valid JWT and attaches the decoded user object to the request), optionalAuth (which attaches the user object if a token is present but continues processing if not), and requireRole (which checks that the authenticated user has a specific role, such as admin). Rate limiting is implemented using a token bucket algorithm with three tiers: generalLimiter (100 requests per 60 seconds for all routes), authLimiter (10 requests per 60 seconds for authentication endpoints), and financialLimiter (5 requests per 1 second for financially sensitive operations such as buy, sell, invest, and withdraw).

The audit logging system maintains a hash chain of all API requests, where each log entry includes the hash of the previous entry, creating a tamper-evident chain similar in concept to a blockchain. Any modification to a historical log entry would break the hash chain, making tampering immediately

detectable. The system also implements double-mint prevention through atomic database UPDATE operations that lock payment order records before processing, preventing concurrent requests from processing the same payment multiple times. Input sanitization removes potentially dangerous content from all request bodies, including HTML tags, JavaScript code, and SQL injection attempts, providing protection against cross-site scripting (XSS) and SQL injection attacks.

CLAIMS

We claim:

1. A computer-implemented method for tokenizing real-world assets on a blockchain platform, the method comprising: receiving asset metadata from an asset owner, the metadata including an asset title, description, category, target raise amount, and listing duration; storing the asset in a database with a draft status; receiving one or more supporting documents for the asset; processing the documents through a multi-stage artificial intelligence pipeline comprising image preprocessing, optical character recognition, text classification, key information extraction, and verification scoring to produce a verification recommendation; transitioning the asset to an active state upon successful verification; receiving investment transactions from a plurality of investors; recording the investment transactions on a custom proof-of-work blockchain as INVEST transaction types; locking investment funds in a per-asset escrow account; and upon the asset reaching its target raise amount, executing a PAYOUT transaction to release escrowed funds to the asset owner.
2. The method of claim 1, wherein the custom proof-of-work blockchain utilizes SHA-256 hashing for block construction and ECDSA secp256k1 elliptic curve cryptography for transaction signing, and supports a plurality of transaction types selected from the group consisting of MINT, TRANSFER, INVEST, DIVEST, PAYOUT, REFUND, FEE, and ASSET_CREATE.
3. The method of claim 1, wherein the multi-stage artificial intelligence pipeline further comprises: applying noise reduction, contrast enhancement, and skew correction to document images in a first stage; extracting raw text from preprocessed images using a deep learning-based optical character recognition engine in a second stage; classifying document type and relevance using natural language processing models in a third stage; extracting structured data points using named entity recognition in a fourth stage; and aggregating outputs from all preceding stages to compute a confidence score and binary recommendation in a fifth stage.
4. The method of claim 1, wherein the asset lifecycle is governed by a deterministic finite state machine with the following state transitions: draft to documents_uploaded to ai_analyzing to verified to compliance_review to active to funding to funded to payout_pending to completed, with rejection branches from ai_analyzing, verified, and compliance_review states leading to a rejected state, and expiration branches from funding and funded states leading to expired, refunding, and closed states.

5. The method of claim 1, further comprising: maintaining a centralized limit-order trading engine for secondary market trading of platform tokens; matching buy and sell orders using a price-time priority discipline where orders are sorted by price level and timestamp; verifying user token balances against the blockchain ledger before order placement; and recording each executed trade as a TRADE transaction on the blockchain.
6. The method of claim 5, wherein the price-time priority matching algorithm sorts buy orders by price in descending order and sell orders by price in ascending order, with ties at the same price level broken by the timestamp of order placement, and wherein the matching process supports both market orders executed at the best available price and limit orders executed at or better than a specified price.
7. The method of claim 1, further comprising computing a dynamic token price using a dual-factor algorithmic formula: $P = P_{\text{initial}} \times (1 + \text{totalSupply}/10000) \times (1 + \text{totalAssetsFunded} \times 0.04)$, where P_{initial} is the base token price, totalSupply is the cumulative number of platform tokens in circulation, and totalAssetsFunded is the cumulative number of assets that have successfully completed their funding period.
8. The method of claim 1, further comprising implementing a multi-tiered security model comprising: JSON Web Token-based authentication with Bearer token scheme; token bucket rate limiting with tiered limits for general, authentication, and financial endpoints; input sanitization middleware that removes potentially malicious content from request bodies; and audit logging with a hash chain integrity verification system where each log entry includes a cryptographic hash of the previous entry.
9. The method of claim 1, further comprising preventing double-minting of payment orders through atomic database UPDATE operations with conditional WHERE clauses that lock the order record before payment verification, ensuring that concurrent requests to process the same order are rejected with zero affected rows.
10. A system for tokenizing real-world assets on a blockchain platform, the system comprising: a processor; a memory coupled to the processor and storing computer-readable instructions that, when executed by the processor, cause the system to perform the method of any one of claims 1 to 9.
11. A non-transitory computer-readable storage medium storing computer-readable instructions that, when executed by a processor, cause the processor to perform the method of any one of claims 1 to 9.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the present invention and, together with the description, serve to explain the principles of the invention.

FIG. 1 illustrates a high-level system architecture diagram showing the five principal layers of the platform: Blockchain Layer, Application Layer, Data Layer, AI Processing Layer, and Security Layer.

FIG. 2 illustrates the transaction structure diagram showing the relationship between a transaction, its ECDSA secp256k1 signature, SHA-256 hash, and type classification.

FIG. 3 illustrates the block structure diagram showing the components of a blockchain block including Merkle root, transactions array, previous hash reference, nonce, and difficulty.

FIG. 4 illustrates the blockchain engine flow diagram showing the progression from transaction pool through proof-of-work mining to chain storage.

FIG. 5 illustrates the asset lifecycle state machine diagram showing all valid state transitions from draft through completed or closed.

FIG. 6 illustrates the trading engine sequence diagram showing the interaction between buyers, sellers, the express API, trading engine, blockchain, and database during order placement and matching.

FIG. 7 illustrates the middleware pipeline diagram showing the sequential processing of HTTP requests through authentication, rate limiting, and audit logging stages.

BEST MODE OF CARRYING OUT THE INVENTION

The best mode currently contemplated by the inventors for carrying out the present invention is implemented as a monolithic Node.js application with the following technical specifications. The application server is built on the Express.js web framework running on the V8 JavaScript engine. The blockchain engine is implemented as a custom module within the application, utilizing the crypto module for ECDSA signature operations and the crypto-js library for SHA-256 hashing. The proof-of-work mining process runs asynchronously within the Node.js event loop, allowing the application to continue processing HTTP requests while mining operations proceed in the background.

The data layer utilizes SQLite as the relational database engine, selected for its zero-configuration deployment, transactional integrity (ACID compliance), and suitability for single-server applications. The database schema comprises nineteen interconnected tables: users, wallets, kyc_verifications, assets, asset_documents, asset_tokens, escrow_accounts, escrow_transactions, coin_orders, coin_trades, economy, price_history, fee_ledger, notifications, payment_gateways, payment_orders, payment_transactions, audit_logs, and sessions. The AI document analysis pipeline utilizes pre-trained machine learning models for OCR, text classification, and named entity recognition, deployed as separate processing modules that communicate with the main application through an asynchronous message queue.

The security layer implements JSON Web Tokens (JWT) for stateless authentication, with token payloads containing the user identifier, role, and KYC verification tier. Rate limiting is implemented using an in-memory token bucket algorithm with configurable parameters for different endpoint categories. The audit logging system writes a sequential log of all API requests to the audit_logs table, with each entry containing the previous entry's hash to form a tamper-evident chain. The system deploys on a single server with a minimum of 2 CPU cores and 4GB RAM, with the blockchain storage file (chain.json) and SQLite

database file stored on persistent disk with regular backup schedules.

INDUSTRIAL APPLICABILITY

The present invention has significant industrial applicability in the financial technology sector, particularly in the emerging field of real-world asset tokenization. The system enables small and medium enterprises, real estate developers, agricultural cooperatives, and other asset owners to access capital markets by tokenizing their assets and offering fractional ownership to a broad base of investors. The AI-powered document verification reduces the cost and time associated with asset due diligence, making the platform accessible to asset owners who cannot afford traditional investment banking services.

The invention also finds application in regulatory compliance contexts, where the immutable blockchain ledger provides a transparent and auditable record of all transactions, escrow operations, and asset lifecycle events. This transparency is particularly valuable in jurisdictions that require detailed record-keeping for financial transactions, anti-money laundering (AML) compliance, and know-your-customer (KYC) verification. The platform's multi-tiered fee structure generates sustainable revenue while maintaining competitive trading costs for users, making it economically viable as a standalone business or as a white-label solution for financial institutions seeking to offer tokenization services to their clients.

Furthermore, the invention's trading engine and algorithmic price discovery model create a self-sustaining economic ecosystem where token value is algorithmically tied to platform activity, providing natural incentives for participation and growth. The custom blockchain infrastructure ensures that the platform can operate independently of public blockchain networks, reducing operational costs, eliminating gas fee dependencies, and providing full control over transaction throughput and confirmation times. These characteristics make the invention particularly suitable for deployment in markets with limited blockchain infrastructure or where regulatory requirements mandate data sovereignty and privacy.